Dictionary:

1.dic is an unordered collection

2.Defined as {key : value}¶

3.key - generally int or str , Unmutable

4.val - any data type , mutable

In [1]:
```
dic1 = dict()  #empty dic 1
dic1
```

Out[1]:
```
{}
```

In [2]:
```
dic2 = {} #empty dic 2
dic2
```

Out[2]:
```
{}
```

In [3]:
```
d1 = {1:4}   # dic with 1 iteam
d1
```

Out[3]:
```
{1: 4}
```

In [4]:
```
d2 = {1:44 , "a" :40.5}  # duc with 2 iteam
d2
```

Out[4]:
```
{1: 44, 'a': 40.5}
```

In [5]:
```

mydict = {'A':'one' , 'B':'two' , 'C':'three'}   # dictionary with character key
mydict
```

Out[5]:
```
{'A': 'one', 'B': 'two', 'C': 'three'}
```

In [6]:
```
mydict = {1:'one' , 'A':'two' , 3:'three'}  # dictionary with mixed keys
mydict
```

Out[6]:
```
{1: 'one', 'A': 'two', 3: 'three'}
```

In [7]:
```
mydict.keys() # returning list of keys
```

Out[7]:
```
dict_keys([1, 'A', 3])
```

In [8]:
```
mydict.values()  # returning list of values
```

Out[8]: dict_values(['one', 'two', 'three'])

In [9]:
```
mydict.items() #returning list of tuple or pair of keys- values
```

Out[9]: dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])

In [10]:
```
mydict = dict([("dad","businessman") ,('mom' ,'housh wife'),("son" , "sunil")])
mydict
```

Out[10]: {'dad': 'businessman', 'mom': 'housh wife', 'son': 'sunil'}

In [11]:
```
keys = {'a' , 'b' , 'c' , 'd'}
mydict2 = dict.fromkeys(keys)     # Create a dictionary from a sequence of keys
mydict2
```

Out[11]: {'c': None, 'd': None, 'b': None, 'a': None}

In [12]:
```
keys = {'a' , 'b' , 'c' , 'd'}
values = (20)
mydict3 = dict.fromkeys(keys,values) # Create a dictionary from a sequence of va
mydict3
```

Out[12]: {'c': 20, 'd': 20, 'b': 20, 'a': 20}

In [13]:
```
keys = {'a' , 'b' , 'c' , 'd'}
value = [10,20,30]
mydict3 = dict.fromkeys(keys,value)
mydict3
```

Out[13]: {'c': [10, 20, 30], 'd': [10, 20, 30], 'b': [10, 20, 30], 'a': [10, 20, 30]}

accessing iteam through key(),get()

In [14]:
```
mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}
mydict
```

Out[14]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}

In [15]:
```
mydict.get(3)
```

Out[15]: 'three'

In [16]:
```
mydict[3]
```

Out[16]: 'three'

In [17]:
```
mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' :'Analyst'}
mydict1
```

Out[17]: {'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}

In [18]:
```
mydict1['job']
```

Out[18]: 'Analyst'

In [19]:
```
mydict1.get('Name')
```

Out[19]: 'Asif'

Add, Remove & Change Items

In [20]:
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hydrabad'}
mydict1
```

Out[20]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hydrabad'}
```

In [21]:
```
mydict1['DOB']= 1995   #changing iteam
mydict1
```

Out[21]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1995, 'Address': 'Hydrabad'}
```

In [22]:
```
mydict1['job']= 'analyst'  #adding iteam
mydict1
```

Out[22]:
```
{'Name': 'Asif',
 'ID': 12345,
 'DOB': 1995,
 'Address': 'Hydrabad',
 'job': 'analyst'}
```

In [23]:
```
mydict1.pop('ID') #removing iteam
mydict1
```

Out[23]:
```
{'Name': 'Asif', 'DOB': 1995, 'Address': 'Hydrabad', 'job': 'analyst'}
```

In [24]:
```
del[mydict1['Address']]   # Removing item using del method
mydict1
```

Out[24]:
```
{'Name': 'Asif', 'DOB': 1995, 'job': 'analyst'}
```

In [25]:
```

mydict1.clear()    # Delete all items of the dictionary using clear method
mydict1
```

Out[25]:
```
{}
```

In [26]:
```
del[mydict1]
mydict1
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[26], line 2
      1 del[mydict1]
----> 2 mydict1

NameError: name 'mydict1' is not defined
```

In [27]:
```
mydict_new = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'korba'}
```

```
mydict_new
```

Out[27]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'korba'}

In [29]:
```
mydict11 = mydict_new.copy()
mydict11
```

Out[29]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'korba'}

In [30]:
```
id(mydict_new)
```

Out[30]: 2201032207296

In [31]:
```
id(mydict11)
```

Out[31]: 2201031980992

In [32]:
```
mydict11['Address']='bsp'
mydict11
```

Out[32]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

In [34]:
```
mydict_new = mydict11
```

In [35]:
```
mydict11
```

Out[35]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

In [36]:
```
mydict_new
```

Out[36]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

In [37]:
```
mydict11['Address']='bsp'
mydict11
```

Out[37]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

In [39]:
```
mydict_new  # mydict_new will be also impacted as it is pointing to the same dic
```

Out[39]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

looping through a dictionary

In [41]:
```
mydict_new
```

Out[41]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}

In [43]:
```
for i in mydict_new:                # Key & value pair
    print(i ,':',mydict_new[i] )
```

```
Name : Asif
ID : 12345
DOB : 1991
Address : bsp
```

In [47]:
```
for i in mydict_new:
    print(mydict_new[i])  # dic value
```

```
Asif
12345
1991
bsp
```

dic membership

In [48]:
```
mydict_new
```

Out[48]:
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'bsp'}
```

In [51]:
```
'Name' in mydict_new   # Test if a key is in a dictionary or not.
```

Out[51]:
```
True
```

In [52]:
```
'Asif' in mydict_new   # Membership test can be only done for keys.
```

Out[52]:
```
False
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: